



ASCAT 2026

7-10 October 2026 | London, UK
Central Hall Westminster

A specialized AI agent for sickle cell patient education in low-resource settings: a benchmarking evaluation

Preliminary results and a call for expert raters

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People already ask chatbots about health

32%

of US adults turned to an AI chatbot for health information in the past year

KFF Health Information and Trust poll, March 2026 (n = 1,343)

40M+

people put healthcare questions to ChatGPT every day

OpenAI, "AI as a Healthcare Ally", January 2026

1,587

user questions in 315 conversations with SickleCellPedia, the SCD-specific slice we can actually see

SCKIN operational data, June 2025 – March 2026

Most people with sickle cell disease live where specialists are scarce, and no benchmark of specialist-level SCD questions compares a purpose-built agent with the general models people actually use.

Who is making sure the answers they receive are actually good?

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SickleCellPedia

SCKIN's free, multilingual SCD assistant

A retrieval-augmented agent grounded exclusively in curated sickle cell sources, with system prompts that prioritise medical accuracy, plain-language accessibility and explicit safety guardrails.

Deployed since 2025 on skin.org and WhatsApp; answers in French, English and other languages. Operational questions from real users informed the question set for this study.

Question for the room: which other SCD tools should sit in this comparison?

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Free

no login, no fee

Web + WhatsApp

where patients already are

Curated sources only

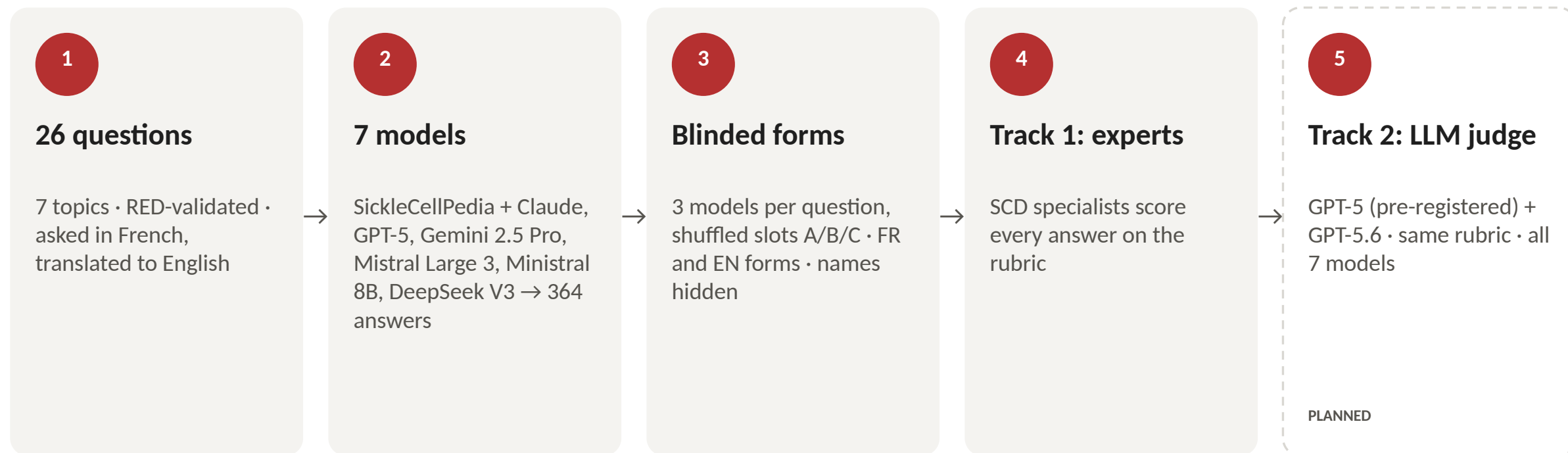
no open-web retrieval

Multilingual

FR · EN · others

Study design

Dual-track blinded evaluation on RED's specialist question set



Rubric Medical accuracy 1–5 (factual correctness, no hallucination, guideline-consistent) · Duty to inform Pass/Fail (safety flag; form wording to be quoted)

Planned vs. available today

The abstract planned ten raters; we are presenting one blinded rating so we can recruit the rest

	Planned (abstract, May 2026)	Available today
Questions	50	26
Models generated / human-rated	7 / 7	7 / 3
Expert raters	10 SCD specialists	1 (Dr Lewis Thomas, EN form)
Expert ratings	≈300	78
Rubric dimensions	Accuracy · Completeness · Ease of understanding · Safety flag	Accuracy · Duty to inform
LLM-as-judge track	GPT-5 on all completions	not yet run

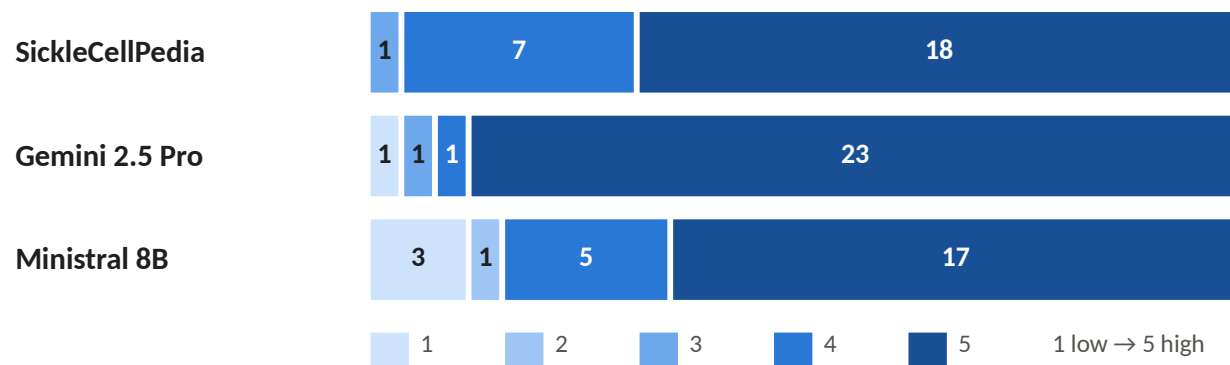
The FR forms (two raters assigned) have no responses yet. Everything on the following slides comes from the one completed EN form.

Preliminary results: 78 blinded ratings

One SCD specialist • Medical accuracy 1–5 • Duty-to-inform Pass/Fail • 26 questions × 3 models

Model	Mean	SD	Scored 5	Scored ≤ 2	Duty fails
SickleCellPedia	4.65	0.55	18 / 26	0	0
Gemini 2.5 Pro	4.73	0.86	23 / 26	1	1
Ministral 8B	4.23	1.34	17 / 26	4	3

Distribution of accuracy scores (n = 26 per model)



Three things the single rating shows

Near-ceiling for everyone. 58 of 78 ratings were 5/5. On these questions the story is not the mean; it is the floor.

Only SickleCellPedia had no floor. No score below 3, zero Duty-to-inform fails. Gemini's mean is higher (4.73 vs 4.65) but it produced one catastrophic answer; Ministral produced three.

Failures cluster on pain. Both pain-management questions produced Fails from general models; travel and pregnancy were near-ceiling for all three.

Every rating, by question and model (1 of 2)

TRANSFUSION

- Q18 Pre-op haemoglobin target before elective surgery
Q19 Exchange vs simple transfusion pre-operatively
Q46 Preventing delayed haemolytic transfusion reaction

PAIN

- Q20 Post-op analgesia without masking an incipient VOC
Q48 IV morphine: titrated vs fixed dose in severe VOC

HYDROXYUREA

- Q29 Hydroxyurea dose for optimal HbF effect
Q30 Laboratory signs of hydroxyurea toxicity
Q32 Non-responder at maximum tolerated dose

PREGNANCY

- Q31 Continue or stop hydroxyurea in pregnancy
Q58 Gestational age for induction of labour
Q59 Severe VOC in the 3rd trimester
Q60 Epidural analgesia in labour
Q65 Fetal monitoring from the 3rd trimester

SickleCellPedia	Gemini 2.5 Pro	Ministral 8B
5	5	5
5	5	1 ✕
5	5	5
5	1 ✕	1 ✕
5	5	1 ✕
5	5	5
4	5	5
3	3	4
5	5	4
5	5	5
5	5	5
4	5	5
5	5	5

Score 1-5, darker = higher · ✕ = Duty-to-inform Fail · one rater · each row is one RED question, each cell the rater's accuracy score for that model's blinded answer.

Every rating, by question and model (2 of 2)

FERTILITY

Q69 Fertility referral before gonadotoxic treatment

Q71 First-line contraception given thrombotic risk

Q74 Counselling a couple who both carry the trait

TRAVEL

Q76 Destinations to formally advise against

Q78 Precautions for high malaria endemicity

Q81 Safe travel plan far from a specialist centre

Q83 Severe VOC abroad without usual centre

Q85 Medical certificate and liaison letter for travel

NEPHROLOGY

Q43 Nephropathy and vital prognosis

Q94 Microalbuminuria screening frequency

Q98 GFR threshold that changes management

Q100 Hydroxyurea dose adjustment in renal impairment

Q103 Should NSAIDs be absolutely avoided

SickleCellPedia	Gemini 2.5 Pro	Ministral 8B
5	5	5
5	5	5
4	5	2
5	5	5
5	5	5
5	5	5
4	5	5
5	5	5
4	5	5
5	5	4
4	5	4
5	4	4
5	5	5

Score 1-5, darker = higher · ✕ = Duty-to-inform Fail · one rater · each row is one RED question, each cell the rater's accuracy score for that model's blinded answer.

Why the general models failed

Specialist shorthand: the questions say "CVO" and never mention sickle cell

RED's questions are written the way French haematologists speak. Q20 asks how to manage post-operative analgesia "sans masquer une CVO débutante"; Q48 asks whether IV morphine should be titrated "dans la CVO sévère". Neither says drépanocytose. The general models guessed what CVO meant. The specialised agent did not have to.

Gemini 2.5 Pro • Q20 • 1, Fail

Read CVO as compartment syndrome ("syndrome des loges") and answered about post-orthopaedic ischaemia.

Ministral 8B • Q20 • 1, Fail

Read CVO as a cerebral vascular complication and advised avoiding opioids so as not to mask neurological signs.

Ministral 8B • Q48 • 1, Fail

Read CVO as a vascular headache and framed morphine titration around giant-cell arteritis.

SickleCellPedia • Q20, Q48 • 5

Answered the sickle cell question: multimodal analgesia, baseline pain documentation, VOC-vs-surgical pain reassessment; titrated morphine within 15–30 min of arrival.

Two readings, both worth discussing: (a) this is exactly how clinicians type, so domain priors are a real safety feature; (b) it is also a question-framing effect that a one-line system prompt might remove. Q19 was different: Ministral conflated exchange transfusion with autologous transfusion. Rater rationale for each Fail is being collected retrospectively.

What this cannot yet tell us

1 One rater, who is a co-author

Blinded, but not independent. No inter-rater reliability can be computed.

2 3 of 7 models seen by a human

SickleCellPedia's own base model (Claude) was not rated, so the retrieval effect is not isolated from the base model.

3 Half the question set, half the rubric

26 of 50 questions; Accuracy and Duty to inform only; no Completeness or Ease-of-understanding scores; no LLM-judge track yet.

4 Ceiling effect

74% of ratings are the maximum score, so the scale separates models mainly through their lowest answers. With a single rater, a significance test would describe one person's scoring rather than specialist opinion, so we report none.

What three or more raters would unlock

The planned analysis, as a pre-registration in effect

Reliability

Krippendorff's α (ordinal) and ICC alongside Fleiss' κ as pre-registered; rater leniency; slot-position (A/B/C) check.

Model comparison

Bootstrap CIs per model; mixed-effects ordinal model with question and rater as random effects; exact binomial CIs on Duty-to-inform failure rates.

Language

FR vs EN rater effects, provided at least one rater completes both forms so language is not confounded with rater.

LLM-judge track

GPT-5 (pre-registered) + GPT-5.6 as a second judge; weighted κ and Spearman vs the human consensus; judge scores for the four models no human has rated (Claude, GPT-5, Mistral Large 3, DeepSeek V3).

Power note: with $SD \approx 0.9$ on the general models, detecting a 0.5-point mean difference at the question level needs roughly 3–4 raters across the 26 questions.

How you can help

Rate the 78 blinded answers. About 30 minutes, in English or French.
Every added SCD specialist turns this talk into a result.

Francophone and LMIC clinicians especially welcome.

Review RED's reference answers or contribute questions from your practice.

Tell us whether Pass/Fail "Duty to inform" is the right safety metric,
and who should rate ease of understanding: clinicians or patients.

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English rating form



Formulaire
d'évaluation (FR)